BIOT6900 Module 5 Review

Sean Cox

Background

In this module, I investigated the various tools which categorize and classify genomic variants. I queried the MyVariantInfo API (BioThings, n.d.) to get information about the DPYD gene (dihydropyrimidine dehydrogenase; UniProt Consortium, n.d.). I then processed and formatted the data so that it could be more readily understood, while being mindful of various pitfalls in the dataset, accounting for this in my data processing pipeline.

MyVariantInfo is an API which collates variant data from multiple sources. For the purpose of this module, I am mainly concerned with the CADD, REVEL, and AlphaMissense scores, which are included in the returned data for DPYD. I first queried this API, and then saved it to a cache for faster processing later on. I then did a quick check of the data to ensure the operation was successful and the data matched what I expected. For DPYD, MyVariantInfo returns a total of 426 identified variants (the veracity of each variant has a defined 'star' evidence level). Of these, CADD was able to identify and label 304, while REVEL and AlphaMissense were able to label 161 and 163, respectively. CADD was able to label variants across various consequence classes (i.e intron variant, stop gain, etc) while REVEL and AlphaMissense only picked up on missense variants. Overall, CADD was able to score 76% of the variants, while REVEL scored 40.2% and AlphaMissense scored 40.8%.

consequence	n	cadd_phred	revel	alphamissense
missense_variant	163	161	161	163
intron_variant	53	44	0	0
synonymous_variant	48	47	0	0
3_prime_UTR_variant	34	11	0	0
stop_gained	30	29	0	0
frameshift_variant	29	0	0	0
splice_donor_variant	17	4	0	0
splice_acceptor_variant	16	2	0	0
splice_region_variant	7	4	0	0
upstream_gene_variant	3	2	0	0
TOTAL	400	304	161	163

percent of the whole variant set scored by each tool

cadd_phred	76.0
revel	40.2
alphamissense	40.8

Interestingly, of the 29 frameshift variants, none of the 3 tools were able to score them.

Thresholds and calibration

Selecting a frequency threshold cutoff is crucial for analyzing this variant data. The frequency of an allele variant in the general population (represented by "`gnomad_af`" in the MyVariantInfo API) is a measure of how often the variant appears at large, at least within the underlying genomic data which was used to label the variants. Variants with a higher frequency are less likely to be pathogenic, so we can safely filter those out and put more focus on the variants which are more likely to cause disease. For example, a filter of 0.001 would remove any variants that occur in more than 0.1% of the measured population. This 0.001 is generally used as a cutoff to limit the variants to those which are present in rare diseases. Manipulating this cutoff is gene specific, however, and I tested various thresholds for DPYD. First, using the standard rare-disease filter value, 69 variants with gnomd_af values for > 0.001 were filtered out. Among these were three variants which the clinvar_sig label (another column which I added to the data – delineating whether the variant is pathogenic, likely pathogenic, benign, etc) labeled as P/LP (Pathogenic or Likely Pathogenic) or being involved in drug response. They are rs3918290 (P/LP, gnomad_af = 0.004) , rs75017182 (drug response, 0.013), and rs1801265 (Pathogenic,

0.27). Even with a cutoff of 0.1 (50 variants removed), rs75017182 is still excluded, so I choose a value of 0.02. This leaves rs1801265 still filtered out, and this variant is an interesting case worthy of discussion. Despite this variant being present in over 27% of the measured population, it is still marked by MyVariantInfo as being pathogenic. The protein encoded by DPYD, dihydropyrimidine dehydrogenase (DPD), is the enzyme responsible for breaking down 80% to 90% of the chemotherapy drugs fluorouracil or capecitabine (ClinPGx, n.d.-a). This variant in particularly may have an affect on the response to these drugs (which is likely why it is marked mistakenly as pathogenic), although more recent evidence has indicated that the effect is not clinically significant, and there are no guidelines recommending dosage changes based on this variant, unlike other DPYD variants (Hamzic et al., 2021). This variant being listed pathogenic is likely an artifact of outdated data, and it can be safely discarded from our variant list. With a cutoff of 0.02, only 38 variants are removed, none of which I consider to be interesting.

For the calibration of the dataset, I choose to use CADD. CADD is suited well for DPYD because it has the best overall coverage as well as the best coverage across consequence classes, of which there are ten identified in DPYD. I calibrated the scores across bands of (0,10], (10, 20], (20, 25], (25, 30] and (30, 50]. These are the ranges of CADD scores present in the data. CADD scores estimate how deleterious a particular variant may be (based on computed evolutionary features), with higher scores predicting a larger impact. Crucially, this does not imply pathogenicity or anything else, it is simply a computed value, and serves only as a hypothesis which indicates a variant is worthy of investigation.

```
cadd_phred calibrated against ClinVar  (P/LP n=39, B/LB n=97)

           n  n_PLP  n_BLB   LR+                evidence
band
(0, 10]   66      0     66  0.00          strong, benign
(10, 20]  24      0     24  0.00          strong, benign
(20, 25]  11      5      6  2.07  no evidence either way
(25, 30]   4      3      1  7.46    moderate, pathogenic
(30, 50]  31     31      0   inf      strong, pathogenic
```

Of the variants which had data that could be used for calibration, most of the bands have enough support for drawing reasonable evidence from. The bands at the ends of the scoring ranges in particular have the strongest support, with the 0-10 bands and 30-50 bands having 66 and 31 examples, respectively. The 10-20 range has 24 examples, which is also sufficient. The lower ranges are given the label of 'strong, benign', indicating confidence that these variants have no effect. The higher score range has a label of 'strong, pathogenic', meaning these variants can be likewise identified with confidence, but are most probably pathogenic. The middle ranges of 20-25 and 25-30 are more dubious, with only 15 total examples across both. These ranges correspond to more unclear evidence about their potential pathogenicity, and their low number of examples further reduces confidence in these bands.

Coverage gaps

There were 29 frameshift variants, and none of the 3 tools were able to score them. These variants occur when a nucleotide is inserted or deleted, misaligning the reading frame. Looking at the AlphaGenome Atlas, it does a pretty good job of classifying some of known high risk variants, scoring 151 variants in total. The variant that it judged to have the highest impact was chr1:g.97549592G>A, which CADD gave a score of 42, indicating that methods agreed that this variant is likely to be highly pathogenic. One variant that CADD, REVEL, and AlphaMissense could not score was chr1:g.97082329C>G, this is a splice donor variant, and the AlphaGenome Atlas was able to score this successfully. This variant landed in the Atlas' top ten highest impact variants for DPYD, hinting that there are potentially high impact variants which systems like CADD and REVEL are blind to. In this case, AlphaGenome was the tool needed to provide an estimation of this variant's pathogenicity. The features which were most prevalent in AlphaGenomes readings were CACTUS_241_WAY (top feature for 51 variants), MERGED_SPLICING and PHASTCONS_470_WAY. CACTUS_241_WAY and PHASTCONS_470_WAY are cross species alignment features, while MERGED_SPLICING is based on the predicted effects of RNA splicing (Google DeepMind, n.d.).

A discordant variant

A very rare variant which I found to have a high discordance between REVEL and AlphaMissense was chr1:g.97450189C>T (rs138616379). This variant scored very high on

REVEL (0.822) but low on AlphaMissense (0.190). Reading more about the variant, it is highly rare, and was only observed 78 times across all populations in the gnomAD, for a total incidence of 0.005% (ClinPGx, n.d.-b). This variant is labeled as VUS (Variant of Unknown Significance) by MyVariantInfo, and there is limited real data on it. I was able to find one study which looked at it and labeled it 'probably damaging' (aligning more with the REVEL assessment) but it made the claim with low statistical significance (Offer et al., 2014). The reason for the discordance is more likely an artifact of the tools than of any underlying biology: the variant is extremely rare, and AlphaMissese and REVEL may have been built on different data in regards to this variant.

Limitations

Continuing to look at chr1:g.97450189C>T, there are various limitations to understanding this variant with the tools I've used. The first limitation with this penetrance: it seems to be an extremely rare variant, but that assumption is derived from population cohorts, and has nothing to do with anything the tools say themselves. The other big limitation is clinical utility. As the scores of various models disagree, there is hardly any clear indication that this variant has a real clinical effect, and even if the models were in agreement, that would not guarantee clinical utility. The only thing that will account for these issues is more data on incidence in the population at large, and deeper investigation into the effects on people who do carry the variant, particularly in their drug response to the chemotherapy drugs which are broken down by DPD. The results of those experiments would determine whether or not this variant is significant or not.

References

BioThings. (n.d.). *MyVariant.info* [Web API]. Retrieved October 25, 2026, from https://myvariant.info/

ClinPGx. (n.d.-a). *[rsID shown on page]* [Variant page]. Retrieved October 25, 2026, from https://www.clinpgx.org/variant/PA166153648

ClinPGx. (n.d.-b). *[rsID shown on page]* [Variant page]. Retrieved October 25, 2026, from https://www.clinpgx.org/variant/PA166153922

Google DeepMind. (n.d.). *AlphaGenome: Using AI to understand the human genome*. Retrieved October 25, 2026, from https://deepmind.google/science/alphagenome/

Hamzic, S., Schärer, D., Offer, S. M., Meulendijks, D., Nakas, C., Diasio, R. B., Fontana, S., Wehrli, M., Schürch, S., Amstutz, U., & Largiadèr, C. R. (2021). Haplotype structure defines effects of common DPYD variants c.85T > C (rs1801265) and c.496A > G (rs2297595) on dihydropyrimidine dehydrogenase activity: Implication for 5‐fluorouracil toxicity. *British Journal of Clinical Pharmacology, 87*(8), 3234–3243. https://doi.org/10.1111/bcp.14742

Offer, S. M., Fossum, C. C., Wegner, N. J., Stuflesser, A. J., Butterfield, G. L., & Diasio, R. B. (2014). Comparative functional analysis of DPYD variants of potential clinical relevance to dihydropyrimidine dehydrogenase activity. *Cancer Research, 74*(9), 2545–2554. https://doi.org/10.1158/0008-5472.CAN-13-2482

UniProt Consortium. (n.d.). *Q12882 · DPYD_HUMAN* [Database entry]. UniProtKB. Retrieved October 25, 2026, from https://www.uniprot.org/uniprotkb/Q12882/entry